

Project Title: Script-Controlled ACL – Restrict Record Access Based on Field Value

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1. Project Overview

This ServiceNow lab implements script-controlled Access Control Lists (ACLs) for the custom u_institution_details table. The main security scenario is branch-based record access: users belonging to the EEE branch should be able to view records where Branch = EEE, while administrators retain full access. The lab also creates separate CREATE, WRITE and DELETE ACLs using custom roles.

2. Project Objective

- ❖ Create an EEE test user and the custom roles bb1, bb2, bb3 and bb4.
- ❖ Create the custom Institution Details table named u_institution_details.
- ❖ Add the required student, faculty, branch and contact fields.
- ❖ Create a READ ACL that restricts access using the Branch value.
- ❖ Create a CREATE ACL using role bb2.
- ❖ Create a WRITE ACL using role bb3.
- ❖ Create a DELETE ACL using role bb4.
- ❖ Test the security behavior using appropriate users and records.
- ❖ Demonstrate record-level protection against unauthorized viewing, modification, creation and deletion.

3. Requirements

3.1 User and Role Requirements

Item	Required Value	Purpose
Test User ID	EEE User	Branch-specific test account
First Name	EEE	Test user
Last Name	User	Test user
Email	eeeuser@gmail.com	Test account email
Custom Role	bb1	READ ACL role
Custom Role	bb2	CREATE ACL role
Custom Role	bb3	WRITE ACL role
Custom Role	bb4	DELETE ACL role

3.2 Table Requirements

Create the following custom table:

- ❖ Label: Institution Details
- ❖ Name: u_institution_details
- ❖ Extends for: false

3.3 Field Requirements

Field Label	Type	Required Configuration
Student Roll Number	Auto Number	Automatically generated
Student Name	Reference	Reference: User
Faculty Name	Reference	Reference: User
Branch	Choice	Choices: ECE, EEE, CSE
Email	String	Text/email value
Phone Number	String	Text/phone value
Description	Multi String	Long/multiple-line text

4. Milestone 1 – Creation of Users and Roles

4.1 Create the EEE User

- ❖ Log in to the ServiceNow instance with admin access.
- ❖ Open User Administration → Users.
- ❖ Click New.
- ❖ Enter User ID: EEE User.
- ❖ Enter First name: EEE.

- ❖ Enter Last name: User.
- ❖ Enter Email: eeeuser@gmail.com.
- ❖ Set the user's branch information to EEE if a Branch field is available in the user record or as required by the lab.
- ❖ Save/Submit the user record.

Screenshot Output:

Figure 1: Creation of EEE User

The screenshot shows the ServiceNow User Administration interface for a user named 'EEE User'. The page is titled 'User - EEE User' and includes a search bar and navigation links. The user details are as follows:

Field	Value
User ID	EEE User
First name	EEE
Last name	User
Title	
Department	
Email	eeeuser@gmail.com
Language	-- None --
Calendar integration	Outlook
Time zone	System (America/Los_Angeles)
Date format	System (yyyy-MM-dd)
Business phone	
Mobile phone	
Photo	Click to add...
Password needs reset	☐
Locked out	☐
Active	☒
Identity type	Human
Internal Integration User	☐

At the bottom of the form, there are buttons for 'Update', 'Set Password', and 'Delete'. Below the form, there is a section for 'Related Links'.

4.2 Create the Custom Roles

- ❖ bb1 – used for READ access
- ❖ bb2 – used for CREATE access
- ❖ bb3 – used for WRITE access
- ❖ bb4 – used for DELETE access

For each role:

- ❖ Navigate to User Administration → Roles.
- ❖ Click New.
- ❖ Enter the role name (bb1, bb2, bb3 or bb4).
- ❖ Enter a short description explaining its purpose.
- ❖ Click Submit.

- ❖ Assign the required roles to the EEE User according to the lab instructions.

Screenshot Output:

Figure 2: Creation of bb1 Role

The screenshot shows the ServiceNow web interface for configuring a role named 'BB1'. The browser tabs include 'ServiceNow Developers' and 'BB1 | Role | ServiceNow'. The URL is 'dev436704.service-now.com/now/nav/ui/classic/params/target/sys_user_role.do%3Fsys_id%3Ddeb8c55e0c3af83106d9cf8ddd4013166%26sysp...'. The page header shows 'Role - BB1' and a search bar. The main form has fields for 'Name' (BB1), 'Application' (Global), and 'Description' (Brief role purpose). There is an 'Elevated privilege' checkbox which is unchecked. Below the form are 'Update' and 'Delete' buttons. A section titled 'Contains Roles' has tabs for 'Applications with Role', 'Modules with Role', and 'Custom Tables'. Below this is a search bar with the text 'for text' and a 'Search' button. A table with the header 'Role = BB1' and a sub-header 'Contains' is shown, but it is empty. The Windows taskbar at the bottom shows the time as 14:25 on 22-09-2026.

5. Milestone 2 – Table Creation

- ❖ In the Application Navigator, search for System Definition → Tables.
- ❖ Click New.
- ❖ Set Label to Institution Details.
- ❖ Set Name to u_institution_details.
- ❖ Set Extends for to false.
- ❖ Save the table.
- ❖ Open the table's Columns/Fields area and add the required fields listed in Section 3.3.
- ❖ Save the table after all fields are created.

Screenshot Output:

Figure 3: Creation of Institution Details Table

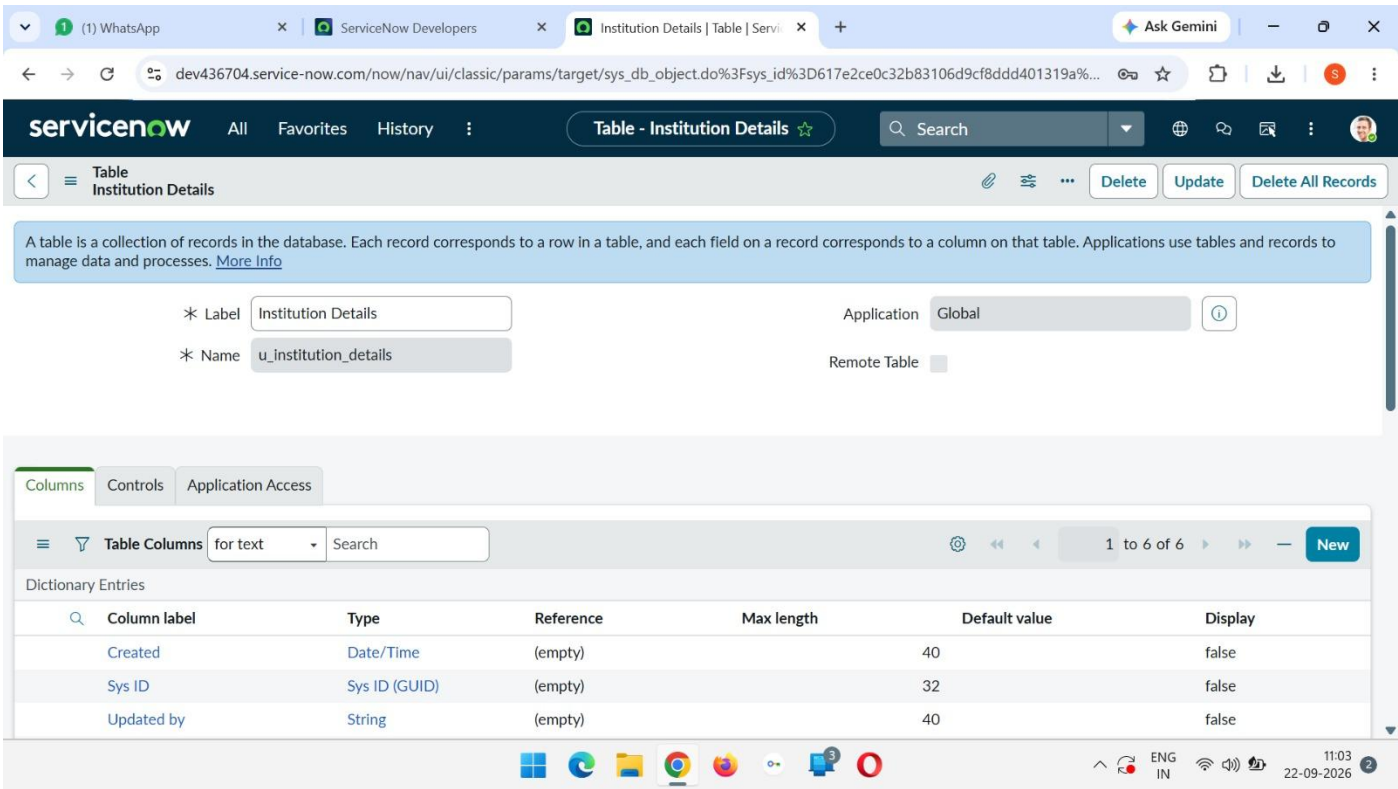


Figure 4: Configuration of Institution Details Table Fields

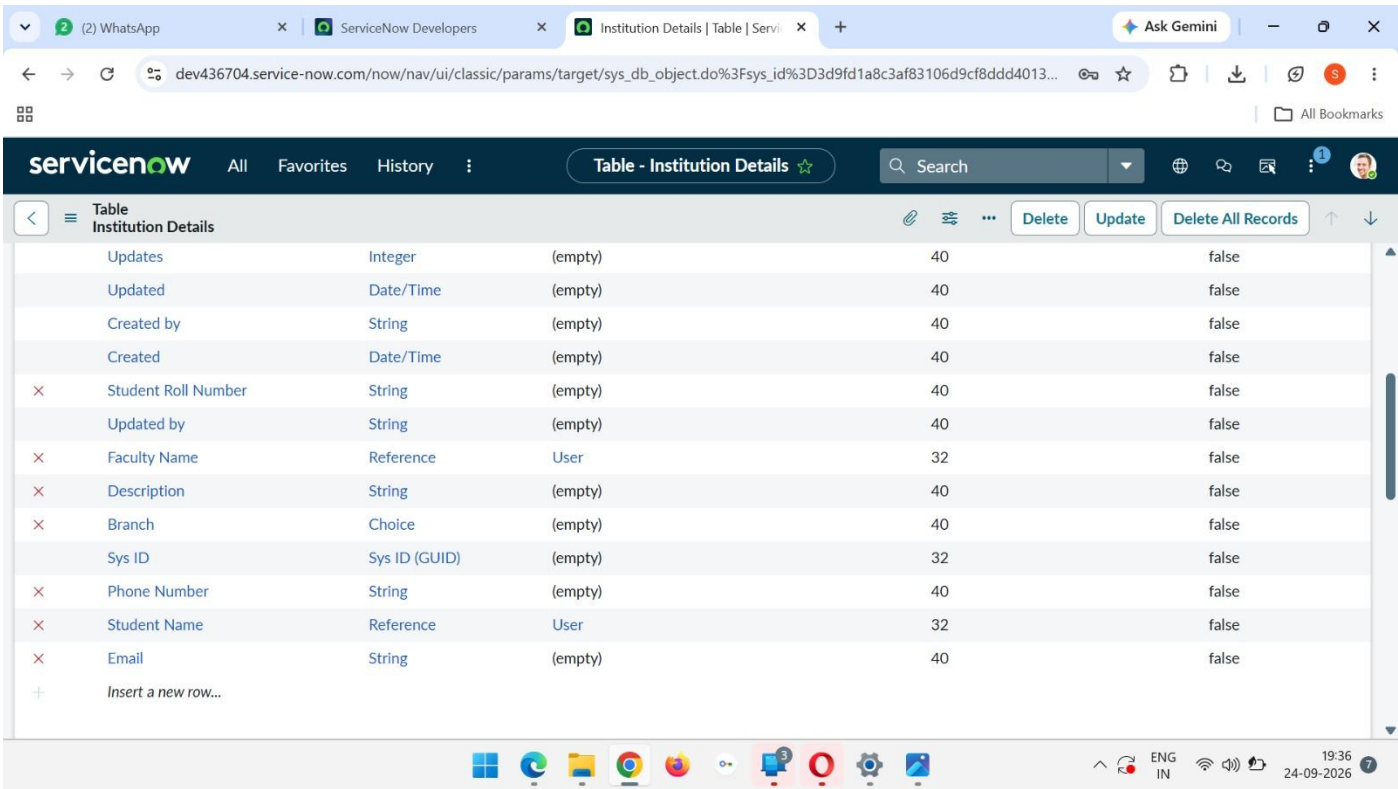


Figure 5: Creation of ECE Branch Record

The screenshot displays the ServiceNow 'Institution Details - Create Created' form. The form is titled 'Institution Details - Create Created' and shows a 'New record' for 'Institution Details'. The form fields are: Student Roll Number (STU100), Faculty Name (Abel Tuter), Student Name (Adela Cervantsz), Branch (ECE), Phone Number, Email, and Description. A 'Submit' button is at the bottom left. The browser address bar shows 'dev436704.service-now.com/now/nav/ui/classic/params/target/u_institution_details.do%3Fsys_id%3D-1'. The Windows taskbar at the bottom shows the date and time as 19:40 on 24-09-2026.

6. Milestone 3 – Create Sample Records

Create enough sample data to test both allowed and denied access.

- ❖ At least one record with Branch = EEE.
- ❖ At least one record with Branch = ECE.
- ❖ At least one record with Branch = CSE.
- ❖ Use valid Student Name and Faculty Name reference values.
- ❖ Enter email, phone and description values where required.

7. Milestone 4 – READ ACL

Purpose: allow the appropriate EEE user to read EEE records while maintaining administrator access.

- ❖ Open System Security → Access Control (ACL).
- ❖ Elevate the security_admin role when required.
- ❖ Click New to create a new ACL.
- ❖ Set Type = record.
- ❖ Set Operation = read.
- ❖ Set Name = u_institution_details.
- ❖ Set Active = true.

- ❖ Set Advanced = true.
- ❖ In the Requires role related list, add bb1.
- ❖ Add the data condition: Branch is EEE.
- ❖ In the Script field, add the following script:

```
(function () {  
  
  // Allow admin users full access  
  
  if (gs.hasRole('admin')) {  
    return true;  
  }  
  
  
  // Allow only EEE branch users to see EEE records  
  
  if (gs.hasRole('bb1')) {  
    return true;  
  }  
  
  
  // Deny access for all others  
  
  return false;  
  
})();
```

- ❖ 12. Click Submit to save the ACL.
- ❖ 13. Save the ACL.

Screenshot Output:

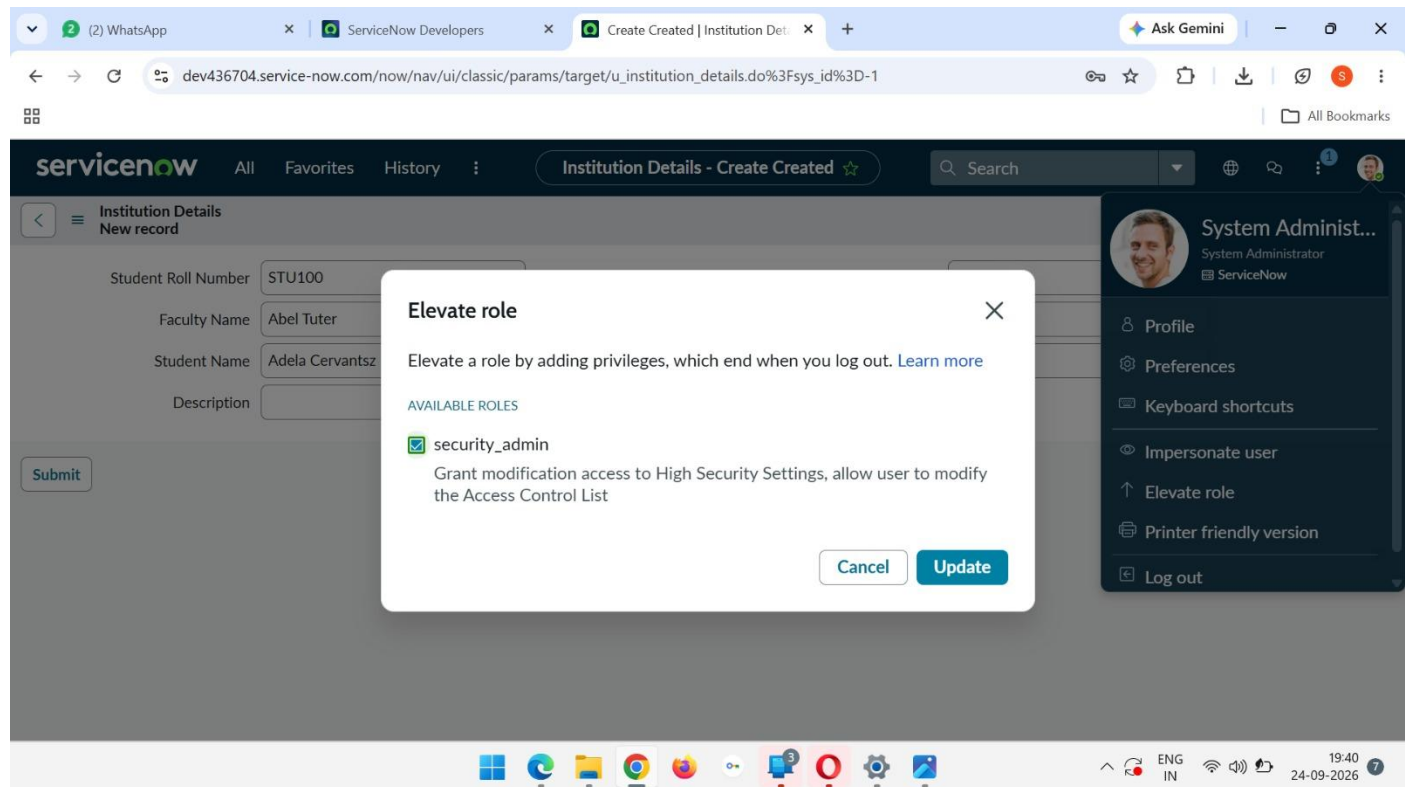


Figure 6: READ ACL Configuration

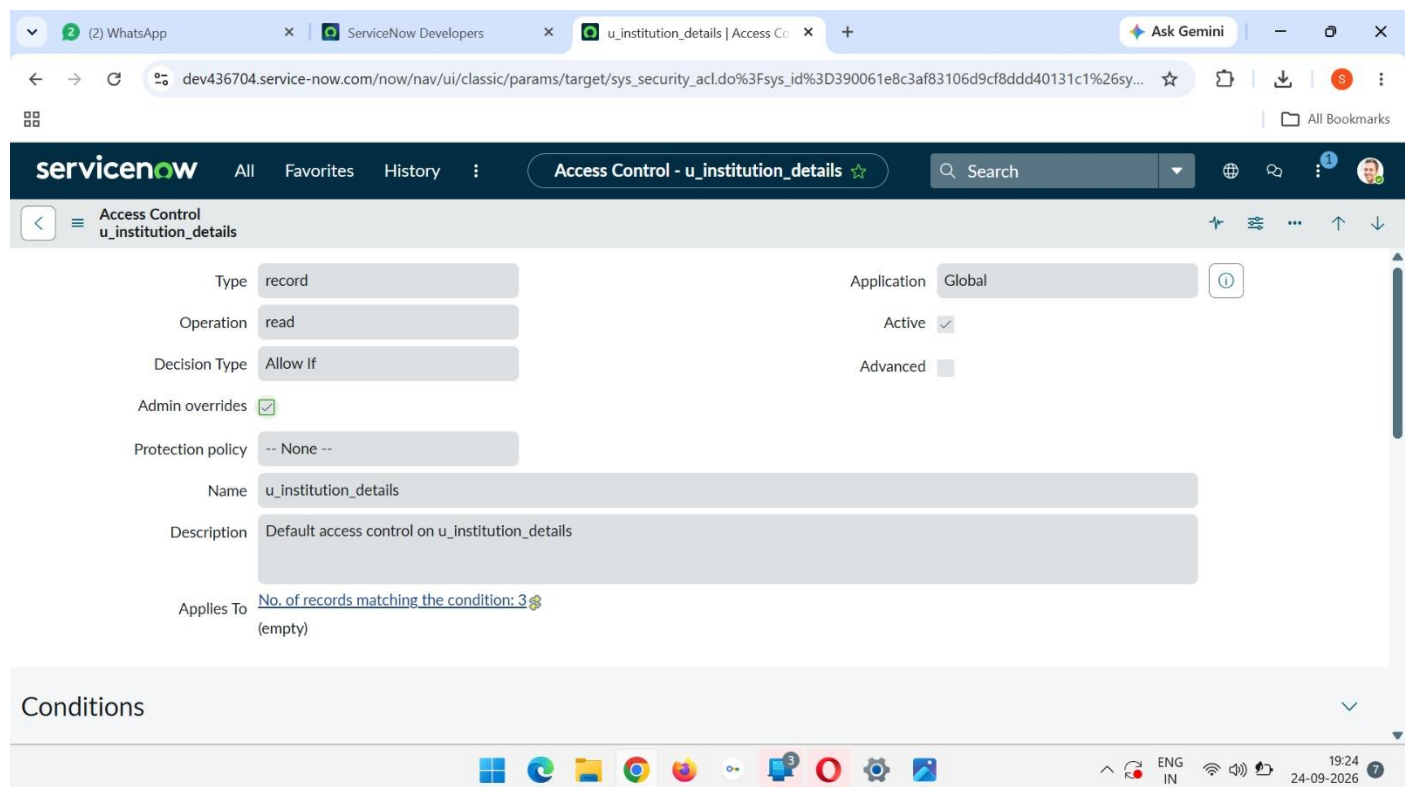


Figure 7: READ ACL – bb1 Role Configuration

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role

BB1

Security Attribute Condition

Local or Existing Existing

Security Attribute

Condition (empty)

Data Condition

Figure 8: READ ACL Script Configuration

Access Control - u_institution_details

Add Filter Condition Add OR Clause

Branch is EEE AND OR X

Advanced Condition

Controlled by References

Script ☒ Turn on ECMAScript 2021 (ES12) mode

```
1 (function () {
2   // Allow admin users full access
3   if (gs.hasRole('admin')) {
4     return true;
5   }
6   // Allow only EEE branch users to see EEE records
7   if (gs.hasRole('bb1')) {
8     return true;
9   }
10  // Deny access for all others
11  return false;
12 })();
```

8. Milestone 5 – CREATE ACL

- ❖ Open the ACL list and click New.
- ❖ Set Type = record.

- ❖ Set Operation = create.
- ❖ Set Name = u_institution_details.
- ❖ Set Active = true.
- ❖ In the Requires role related list, add bb2.
- ❖ No Data condition is required according to the supplied lab instructions.
- ❖ Save the ACL record.

Screenshot Output:

Figure 9: CREATE ACL Configuration

The screenshot shows the ServiceNow interface for configuring an Access Control List (ACL) for the 'u_institution_details' table. The browser address bar shows the URL: dev436704.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3D9e2db460c3ab83106d9cf8ddd40131e8%26s... The ServiceNow header includes the logo, navigation tabs (All, Favorites, History), and a search bar. The main form is titled 'Access Control - u_institution_details' and includes the following fields:

- Type:** record
- Operation:** create (highlighted with a green border)
- Decision Type:** Allow If
- Application:** Global
- Active:** ☒
- Advanced:** ☒
- Admin overrides:** ☒
- Protection policy:** -- None --
- Name:** Institution Details [u_institution_details] (selected from a dropdown)
- Description:** (empty text field)
- Applies To:** No. of records matching the condition: 3 (with a link icon)
- Buttons:** Add Filter Condition, Add OR Clause
- Condition Builder:** -- choose field --, -- oper --, -- value --

At the bottom of the form, there is a section labeled 'Conditions' with a dropdown arrow. The Windows taskbar at the bottom shows the Start button, task view, and several application icons (Edge, File Explorer, Chrome, Firefox, etc.). The system tray on the right shows the date and time: 22-09-2026, 12:09.

Figure 10: CREATE ACL – bb2 Role Configuration

The screenshot shows the ServiceNow web interface for configuring an Access Control Rule. The browser tabs include 'WhatsApp', 'ServiceNow Developers', and 'u_institution_details | Access Co...'. The URL is 'dev436704.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3Db0052564c3ef83106d9cf8ddd4013163%26sys...'. The page title is 'Access Control - u_institution_details'. A search bar is visible. A blue information box states: 'Access Control Rules have two decision types, and these types will behave differently depending on conditions. 1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access. 2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access. More Info'. Below this, the 'Requires role' section shows a table with one entry: 'Role' and 'BB2'. The 'Security Attribute Condition' section has a dropdown for 'Local or Existing' set to 'Existing', and a 'Security Attribute' field. The 'Data Condition' section is empty. The bottom of the screen shows a Windows taskbar with various icons and a system tray with the date '24-09-2026' and time '19:23'.

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role
BB2

Security Attribute Condition

Local or Existing: Existing

Security Attribute:

Condition (empty)

Data Condition

9. Milestone 6 – WRITE ACL

- ❖ Click New to create the Write ACL.
- ❖ Set Type = record.
- ❖ Set Operation = Write.
- ❖ Set Name = u_institution_details.
- ❖ Set Active = true.
- ❖ In the Requires role related list, add bb3.
- ❖ No Data condition is required according to the supplied lab instructions.
- ❖ Save the ACL record.

Screenshot Output:

Figure 11: WRITE ACL Configuration

The screenshot shows the ServiceNow interface for configuring an Access Control rule. The browser tabs include WhatsApp, ServiceNow Developers, and the current page 'u_institution_details | Access Co'. The URL is 'dev436704.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3Da8f565e4c3ef83106d9cf8ddd40131dc%26sy...'. The ServiceNow header shows 'Access Control - u_institution_details' and a search bar. The configuration form includes the following fields:

- Type: record
- Operation: write
- Decision Type: Allow If
- Admin overrides: ☒
- Protection policy: -- None --
- Name: u_institution_details
- Description: (empty)
- Applies To: No. of records matching the condition: 3 (empty)
- Application: Global
- Active: ☒
- Advanced: ☐

Below the form is a 'Conditions' section. The Windows taskbar at the bottom shows the time as 19:25 on 24-09-2026.

Figure 12: WRITE ACL – bb3 Role Configuration

This screenshot shows the 'Conditions' section of the ServiceNow Access Control configuration. It includes a blue informational box about decision types and a table for roles.

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role
Role ▲
BB3

Below the table is the 'Security Attribute Condition' section with the following fields:

- Local or Existing: Existing
- Security Attribute: (empty)

The Windows taskbar at the bottom shows the time as 19:26 on 24-09-2026.

10. Milestone 7 – DELETE ACL

- ❖ Click New to create the Delete ACL.
- ❖ Set Type = record.
- ❖ Set Operation = Delete.
- ❖ Set Name = u_institution_details.
- ❖ Set Active = true.
- ❖ In the Requires role related list, add bb4.
- ❖ No Data condition is required according to the supplied lab instructions.
- ❖ Save the ACL record.

Screenshot Output:

Figure 13: DELETE ACL Configuration

The screenshot shows the ServiceNow web interface for configuring an Access Control List (ACL) for the 'u_institution_details' table. The browser tabs include 'WhatsApp', 'ServiceNow Developers', and 'u_institution_details | Access Co'. The URL is 'dev436704.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3D06462128c3ef83106d9cf8ddd40131f5%26sy...'. The ServiceNow header shows 'Access Control - u_institution_details' and a search bar. The ACL configuration form includes the following fields:

- Type: record
- Operation: delete
- Decision Type: Allow If
- Application: Global
- Active: ☒
- Advanced: ☐
- Admin overrides: ☒
- Protection policy: -- None --
- Name: u_institution_details
- Description: (empty text area)
- Applies To: [No. of records matching the condition: 3](#) (empty)

Below the form is a 'Conditions' section with a dropdown arrow. The Windows taskbar at the bottom shows the date and time as 19:20 on 24-09-2026.

Figure 14: DELETE ACL – bb4 Role Configuration

dev436704.service-now.com/now/nav/ui/classic/params/target/sys_security_acl.do%3Fsys_id%3D06462128c3ef83106d9cf8ddd40131f5%26sy...

servicenow All Favorites History Access Control - u_institution_details Search

Access Control u_institution_details

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role 1 to 1 of 1

Role

BB4

Security Attribute Condition

Local or Existing Existing

Security Attribute

Condition (empty)

11. ACL Configuration Summary

Operation	ACL	Type	Role	Condition
Read	READ	Record	bb1	Branch is EEE + script-controlled branch check
Create	CREATE	Record	bb2	None
Write	WRITE	Record	bb3	None
Delete	DELETE	Record	bb4	None

12. Testing Requirements

Test	Action	Expected Result
Admin – READ	Open EEE and non-EEE records	Admin can access protected records.
EEE User – READ EEE	Open an EEE record	Access is allowed.
EEE User – READ non-EEE	Open ECE/CSE record	Access is denied when the branch rule applies.
bb2 – CREATE	Create an Institution Details record	Creation is allowed for a user with bb2.

Without bb2 – CREATE	Attempt to create a record	Creation is denied if no other applicable ACL grants access.
bb3 – WRITE	Edit an existing record	Update is allowed for a user with bb3.
Without bb3 – WRITE	Attempt to edit	Update is denied if no other applicable ACL grants access.
bb4 – DELETE	Delete an existing record	Deletion is allowed for a user with bb4.
Without bb4 – DELETE	Attempt to delete	Deletion is denied if no other applicable ACL grants access.

13. Important Security Notes

- ❖ Do not give the admin role to the EEE test user when performing negative ACL tests; administrator access can bypass normal restrictions.
- ❖ Use separate sessions or an incognito/private browser window when switching between admin and test-user accounts.
- ❖ ACLs are evaluated server-side. Hiding a button or form element is not a replacement for an ACL.
- ❖ Existing table, field or parent ACLs can affect the final result. If a test behaves unexpectedly, review all applicable ACLs.
- ❖ Use the exact field names generated by the instance. The custom table field may have a name different from its display label.
- ❖ The READ ACL shown in the supplied instructions contains both a role/data condition and an Advanced option. Follow the lab's required configuration exactly.

14. Evidence / Screenshot Requirements

- ❖ EEE User record showing the created user.
- ❖ Roles bb1, bb2, bb3 and bb4.
- ❖ Institution Details table configuration.
- ❖ Table fields/columns showing all required fields.
- ❖ READ ACL configuration.
- ❖ CREATE ACL configuration.
- ❖ WRITE ACL configuration.
- ❖ DELETE ACL configuration.
- ❖ EEE record and non-EEE records used for testing.

- ❖ Successful and denied access results.
- ❖ Final lab/task completion status.

15. Completion Checklist

- ❖ EEE User created.
- ❖ bb1, bb2, bb3 and bb4 created.
- ❖ Roles assigned as required.
- ❖ u_institution_details table created.
- ❖ Student Roll Number configured as Auto Number.
- ❖ Student Name configured as Reference → User.
- ❖ Faculty Name configured as Reference → User.
- ❖ Branch configured as Choice: ECE, EEE, CSE.
- ❖ Email configured as String.
- ❖ Phone Number configured as String.
- ❖ Description configured as Multi String.
- ❖ READ ACL created.
- ❖ CREATE ACL created.
- ❖ WRITE ACL created.
- ❖ DELETE ACL created.
- ❖ EEE branch read restriction tested.
- ❖ Admin access tested.
- ❖ Create/Write/Delete permissions tested.
- ❖ Screenshots captured.
- ❖ Lab shows Completed/Passed.

16. Expected Outcome

By completing this lab, the learner demonstrates an understanding of ServiceNow record-level security using ACLs. READ access is controlled using the EEE branch rule, while CREATE, WRITE and DELETE access are separated through the custom roles bb2, bb3 and bb4. The resulting configuration protects records from unauthorized viewing, creation, modification and deletion across ServiceNow forms, lists and applicable interfaces.

Screenshot Output:

Figure 15: Final ACL Testing Result:

ServiceNow DevelopersInstitution Details | ServiceNow

dev436704.service-now.com/now/nav/ui/classic/params/target/u_institution_details_list.do%3Fsysparm_clear_stack%3Dtrue

Ask Gemini

servicenow

AllFavoritesHistory

Institution Details

Branch

Search

Actions on selected rows...

New

All

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE			Creator User		Alene Rabeck	STU200

1 to 1 of 1

WindowsTaskbarIconsSystemTray